JALEN VUONG

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EDUCATION

B.S. in Mechanical Engineering, Rice University — Expected 2028
B.A. in Classical Studies, Rice University — Expected 2028
GPA: 3.9

SKILLS

Technical Skills Fusion 360, Onshape, 3D Printing, MIG/Arc Welding, Arduino, Python, MATLAB, FEA

EXPERIENCE

Research Assistant — Aug 2026 – Present
CARA Lab — *Houston, TX*

- Researched state estimation methods for small-scale autonomous vehicle
- Developed position and orientation estimation algorithms from LiDAR data using particle filters
- Worked with multidisciplinary team to develop competition vehicle for collegiate autonomous racing

Summer Intern — May 2026 – June 2026
KourtLit — *Houston, TX*

- Customized portable projector given space and portability constraints
- Collaborated on design requirements and components across different subteams
- Created custom parts using CAD, practiced documentation methods for easier communication

Summer Intern — Jul 2025 – Aug 2025
Haast Autonomous — *Houston, TX*

- Developed solutions to reduce vibrational damage to payload inside eVTOL transport
- Designed drone models in Fusion 360 with parametric design with other interns
- Researched airfoil and airframe designs for future prototypes

Workshop Assistant — Oct 2025 – Present
Moody Center for the Arts — *Houston, TX*

- Managed workshop space for various projects and hosted student work session for 3D printing final
- Taught peers how to use rapid prototyping equipment, such as 3D printers, lasercutters, and more
- Aided Rice faculty in professional projects like using a CNC for wooden props or metal engraving

PROJECTS

- **Inverted Pendulum Control:** Designed an inverted pendulum balanced on a cart using Arduino in CAD. Calculated Lagrangian equations of motion and simulated in Python with LQR calculation tools.
- **PID Control for RC Car:** Coded simple PID loop to keep RC car moving in a straight line. Optimized constants to meet self-defined criteria for distance and drift.
- **Manual Transmission Gearbox CAD:** Modeled a simplified manual transmission gearbox using Fusion 360. Practiced good CAD workflow techniques, and utilized joints in assembly design.
- **Portable Changing Table Design:** Created an end-to-end solution to a client problem for EDES 220 course, in a team. Presented final solution to panel of clients and professors.

EXTRACURRICULAR ACTIVITIES

- **Rice Electric Vehicle:** Engineered components for Shell Eco-Marathon competition vehicle.
- **Rice Racing:** Optimized and designed steering and braking subsystems for Formula SAE competition vehicles.
- **Doerr Institute:** Attended leadership training, identified strengths and created actionable goals.
- **Club Sports:** Active participant in Rice University Club Baseball and Club Swim teams.